

Set 28: Equations and observations

An important skill in chemistry is the ability to predict the products when given some reactants. This requires knowledge of the general reaction types. From the products, an equation can be written and the observations expected can be described. Observations include the dissolving or formation of solids (with colours), evolution of gases (including colours and odour) and changes in solution colour. They may also include the formation or absorption of heat and even the production of light and sound.

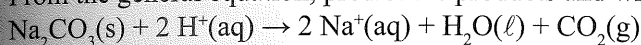
Example

Solid sodium carbonate is added to dilute hydrochloric acid solution.

Step 1 Identify the type of reaction.

This is an acid + carbonate reaction and has the general equation
Acid + carbonate $\rightarrow$ salt + water + carbon dioxide

Step 2 From the general equation, predict the products and write the ionic equation.



Step 3 Describe the observations expected

White solid dissolves; colourless, odourless gas evolved; colourless solution formed.

Set 28: Exercises

For the following reactants, write balanced equations and describe the expected observations.

1. Marble chips (calcium carbonate) are added to dilute nitric acid.
2. Magnesium metal is added to dilute hydrochloric acid.
3. Baking soda (sodium hydrogencarbonate) is added to vinegar (acetic acid).
4. Sodium hydroxide solution is added to sulfuric acid solution.
5. Solid sodium hydroxide is added to dilute hydrochloric acid.
6. Dilute sulfuric acid is added to solid cobalt carbonate.
7. Barium chloride solution is added to dilute sulfuric acid.
8. Lead(II) nitrate solution is added to sodium iodide solution.
9. Gold(III) chloride solution is added to solid copper.
10. Sodium metal is added to water.
11. Potassium carbonate solid is added to an excess of dilute nitric acid.
12. Potassium hydroxide solution is added to nitric acid solution.
13. Silver nitrate solution is added to sodium bromide solution.
14. Nickel metal is added to copper nitrate solution.
15. Iron(II) sulfate solution is added to potassium hydroxide solution.

Notes